

AI Workflow & Implementation Notes

Converting a High-Fidelity Prototype into a Shopify Online Store 2.0 Theme — Purelane Project

1. Goal & Requirements

- **Objective:** Convert a high-fidelity HTML/CSS prototype (purelane-homepage.html) into a fully modular, performant, and editable Shopify Online Store 2.0 Theme.
- **Key Focus:** 100% pixel-perfect visual fidelity, smooth animations (60 FPS), and dynamic merchant controls via the Shopify Theme Editor.

2. Step-by-Step Workflow

Step 1: Codebase Analysis & Section Architecture

- Deconstructed the static prototype into independent, reusable Liquid sections:
 - purelane-scenes.liquid — animated water background & sunlit caustics
 - purelane-ticker.liquid — continuous looping announcement bar
 - purelane-header.liquid — floating glassmorphic navigation pill
 - purelane-hero.liquid — hero section with bottle carousel & trust badges
 - purelane-reviews.liquid, purelane-ingredients.liquid, purelane-proof.liquid
 - purelane-combos.liquid, purelane-bundles.liquid, purelane-shop.liquid — bestsellers grid
 - purelane-range.liquid, purelane-signup.liquid, purelane-footer.liquid
- Configured section schemas ({% schema %}) and block presets in templates/index.json and sections/header-group.json for full customization in the Shopify Customizer.

Step 2: Visual Styling & Cinematics

- **Mint Water Theme:** Recreated multi-layered background scenes with SVG filters (feTurbulence, feDisplacementMap), caustic light beams, rising bubbles, and light vignettes.
- **Glassmorphic UI:** Implemented frosted glass cards using backdrop-filter, subtle gradient borders, and tailored mint/teal/gold color palettes.
- **Responsive Layout:** Balanced vertical alignment and padding for consistent presentation across desktop, tablet, and mobile viewports.

Step 3: Interactivity & Performance Engineering

- **Progress Rail Synchronization:** Replaced unreliable scrollTop loops with an accurate getBoundingClientRect() calculation so vertical progress dots activate step-by-step as sections enter the viewport.
- **Ultra-Fast Scroll Rendering:**
 - Configured IntersectionObserver with an eager rootMargin of 300px and a 0.01 threshold so content renders before reaching the screen.
 - Replaced heavy blur filters with snappy, hardware-accelerated CSS translate3d transitions (0.25s) for smooth 60 FPS scrolling.
- **Smart Header Behavior:** Ticker sits at the top and scrolls away naturally (position: absolute), while the floating navbar smoothly transitions to a compact top-docked pill on scroll.

- **Seamless CTAs:** Connected "Shop Now" and "How It Works" hero buttons to smooth-scroll directly to #shop and #proof.

Step 4: Continuous Deployment & Verification

- Pushed all incremental updates live to the Shopify store using Shopify CLI (shopify theme push --allow-live).
- Committed every feature and fix to Git for version control and clean tracking.

3. Key Takeaways

- **Modularity:** Every section is 100% customizable from the Shopify Admin UI.
- **Performance:** Clean CSS transitions and eager JavaScript observers eliminate render lag during scroll.
- **Design Accuracy:** Preserved the exact typography, water animations, and glassmorphic look of the original prototype.